

Due on tuesday, February 7th, in class

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The relativistic factor

$$\gamma(v) = \frac{1}{\sqrt{1 - (v/c)^2}}$$

is central in special relativity. It is a function of the speed (magnitude of velocity) v . It's vital to know the properties of $\gamma(v)$ in some detail.

If tired of carrying along factors of c , you could use the common notation $\beta = \frac{v}{c}$. If there are multiple speeds involved, such as u and v , you could use $\beta_u = \frac{u}{c}$ and $\beta_v = \frac{v}{c}$. But please express final results in terms of u , v and c again.

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1. (a) [2 pts.] Show that

$$\frac{1}{\gamma^2} + \frac{v^2}{c^2} = 1 \quad \left\{ \begin{array}{l} \text{Here } \gamma \text{ is used as} \\ \text{shorthand for } \gamma(v) \end{array} \right.$$

- (b) [2 pts.] Show that

$$\gamma(v) \times (c + v) = c \sqrt{\frac{c + v}{c - v}}$$

- (c) [6 pts.] Show that

$$\frac{d\gamma}{dt} = \frac{1}{c^2} \gamma^3 v \dot{v} \quad \text{and} \quad \frac{d\gamma}{dt} = \frac{1}{c^2} \gamma^3 \vec{v} \cdot \vec{a}$$

where we use the shorthands $\gamma = \gamma(v)$ and $\dot{v} = \frac{dv}{dt}$. Note that $\dot{v}$ is the derivative of the speed, not the derivative of the 3-vector $\vec{v}$, i.e., $\dot{v}$ is not the acceleration vector $\vec{a} = \frac{d\vec{v}}{dt}$.

- (d) [5 pts.] If $w = (u + v)/(1 + uv/c^2)$, derive a relation of the form

$$\gamma(w) = \gamma(u)\gamma(v) \times \text{something}$$

and state explicitly what this something is.

2. **[3 pts.]** Present a neat hand-drawn plot of $\gamma(v)$ as a function of v or v/c . (As usual, you can use a computer plotting programme to figure out what this looks like, but please submit a hand-drawn plot.)

Looking at your curve, I should be able to immediately see the value of $\gamma(v)$ at zero speed, the slope at zero speed, and at which speed, if any, $\gamma(v)$ diverges.

3. The binomial series

$$(1+x)^\epsilon = 1 + \epsilon x + \frac{1}{2!}\epsilon(\epsilon-1)x^2 + \frac{1}{3!}\epsilon(\epsilon-1)(\epsilon-2)x^3 + \dots$$

holds for all $|x| < 1$. Please find a way to remember this, so that you can use it without having to look it up. (Look up ‘binomial series’ online.)

- (a) **[3 pts.]** Derive the binomial expansion using the Taylor expansion. State clearly which function you are expanding and around what point.
 - (b) **[2 pts.]** Expand $(P+Q)^\epsilon$, first assuming $|Q| < |P|$, and then for the case $|P| < |Q|$.
 - (c) **[3 pts.]** Use the binomial expansion to find good approximations for $(1001)^{1/3}$ and $(15)^{1/4}$, without using a calculator.
 - (d) **[SELF]** For which values of ϵ does the series terminate after a finite number of terms?
 - (e) **[4 pts.]** Expand γ in powers of v/c , including at least up to order $(v/c)^6$. This expansion is useful for small v , when γ is close to 1.
 - (f) **[3 pts.]** If you truncate the series, you get an approximant for $\gamma(v)$. Plot $\gamma(v)$ as a function of v/c , together with the approximants obtained by keeping the first one, two, three or four terms of the series. This should show how successive approximants do a better and better job of approximating our $\gamma(v)$ function.
Please submit only a neat hand-drawn plot, no computer printouts.
 - (g) **[4 pts.]** Express v as a function of $1/\gamma$. Expand using the binomial expansion, at least up to order $1/\gamma^6$. For which values of v is this expansion useful?
4. **[3 pts.]** A low-flying earth satellite travels at about 8000 m/s. What is the factor γ for this speed?

5. We will be dealing a lot with *hyperbolic functions*.

- (a) **[3 pts.]** Look up (or recall) and report the definitions of the hyperbolic functions $\cosh(x)$, $\sinh(x)$ and $\tanh(x)$, in terms of the exponential functions e^x and e^{-x} .
- (b) **[3 pts.]** Provide neat hand-drawn plots of the three hyperbolic functions.
(You can use a computer program if you want to figure out how the functions look like, but you really should be able to figure out how these functions look like, from the definitions in terms of e^x and e^{-x} .)
- (c) **[4 pts.]** Express the corresponding three trigonometric functions in terms of e^{ix} and e^{-ix} . Are the relations

$$\sinh(x) = c_1 \sin(ix) \quad \text{and} \quad \sinh(ix) = c_2 \sin(x)$$

correct for some value of c_1 and c_2 ? Which values?